

Student Research Internship · Team Forelle

IoT & Sensor Systems Engineer

FunkyForelle | Deploying River Sensors to Protect Swiss Fish

About FunkyForelle

FunkyForelle is an open-source project building an AI early warning system for Swiss fisheries managers. Our model predicts dissolved oxygen (DO) — the variable that kills fish during summer heatwaves — across Swiss rivers. To validate the model and create the first real-time DO data stream for a small Swiss river, we need a physical sensor network deployed in the field.

The project is funded by the Prototype Fund Switzerland (CHF 60,000). You will be based at OST Rapperswil — chosen because the campus sits on Lake Zurich, giving you immediate access to a water body for sensor testing from day one.

Your Role — Team Forelle

You are the sole IoT engineer on Team Forelle. You will design, procure, calibrate, and deploy a network of sensors across 7–8 Swiss river sites — ranging from a full multi-parameter station at Patrick Hager's pilot river to simpler temperature + depth loggers at validation sites.

- **Month 1** — Research & procurement: Survey sensor technologies used by Swiss monitoring agencies (BAFU, cantonal offices, Eawag). Market analysis of DO, temperature, and depth sensors. Select and order hardware. Begin bench testing at OST campus (Lake Zurich).
- **Month 2** — Calibration & site preparation: Calibrate sensors against reference measurements. Confirm 7–8 field sites with Patrick Hager. Prepare mounting, weatherproofing, and telemetry (GSM/LoRa).
- **Month 3** — Field deployment: Install all sensor stations. Commission real-time data transmission to the shared pipeline. Deliver live data stream to Team Funky.
- **Month 4** — Reserve: Troubleshooting, system hardening, sensor maintenance guide, open-source documentation.

What You Will Deploy

- Pilot site (Patrick's river): In-Situ Aqua TROLL 500 — DO + temp + conductivity + depth in one unit
- 3 secondary sites: YSI ProODO — DO + temperature, with GSM telemetry
- 3 validation sites: Keller Series 36W — temperature + depth loggers
- Telemetry: GSM modules transmitting every 10 minutes to shared data pipeline
- Total: 7 sites, real-time data across a range of Swiss river types

What You Will Work With

- Hardware: DO probes, multi-parameter sondes, depth/temp loggers, GSM telemetry modules
- OST campus: IoT lab, calibration equipment, 3D printer (enclosures), Lake Zurich for bench testing — all at no cost
- Reference networks: BAFU FOEN stations, Eawag deployments — for context and calibration
- Expert advisor: Patrick Hager (Fischereiverein See + Gaster) — defines pilot site, advises on field conditions
- Weekly check-ins with project leads Olga and Tatyana; data handover to Team Funky

What We Are Looking For

Required:

- Enrolled in BSc or MSc in Electrical Engineering, Computer Engineering, Environmental Engineering, IoT/Embedded Systems or related field
- Hands-on experience with microcontrollers, sensors, or embedded systems
- Interest in field work and real environmental data
- Comfortable working independently in the field

Nice to have:

- Experience with water quality sensors or environmental monitoring hardware
- Knowledge of GSM, LoRa, NB-IoT or MQTT telemetry protocols
- Python for data ingestion pipelines
- Interest in open-source hardware

What We Offer

- CHF 2,000/month — 4 months (~15–20 hrs/week)
- Academic credit: we will support thesis or project course connection with your university
- OST Rapperswil campus and Lake Zurich for field testing from day one
- CHF 22,000 hardware budget — you specify, select, and buy the sensors
- Real field deployment: your sensors stream live data to a production AI model
- Expert mentorship from Patrick Hager and project leads
- Open-source contribution: sensor selection report and firmware published

Logistics

- Duration: 4 months — November 2026 to February 2027
- Location: OST Rapperswil campus + river field sites (Canton St. Gallen / Zurich area) + remote
- Start date: Flexible, ideally 1 November 2026
- OST students especially welcome — local campus access and academic credit alignment

How to Apply

Email polar.bear.before.lunch@gmail.com (one page max):

1. Your current program and year of study
2. Two or three sentences on why this project interests you
3. Any relevant hardware projects, lab work, or field experience

We respond to all applications within one week.